

# Carbon Composition Resistors in Guitar Amplifiers



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In the tweed era, guitar amplifiers were built almost exclusively with carbon composition resistors. For sonic authenticity they continue to be built into many amps today, despite concerns over noise and reliability.

*Sure, they are a little noisier. So what. The noise floor disappears the minute you strum the first chord. Occasionally one of them will get sputtery, but you just have to fix it. By virtue of the fact that they are built so service-tech friendly, should you have a component that goes south - on that rare occasion where the 100k plate resistors get sputtery on you - it's a five-minute fix.*<sup>1</sup> - Mark Baier, Victory Amp Company

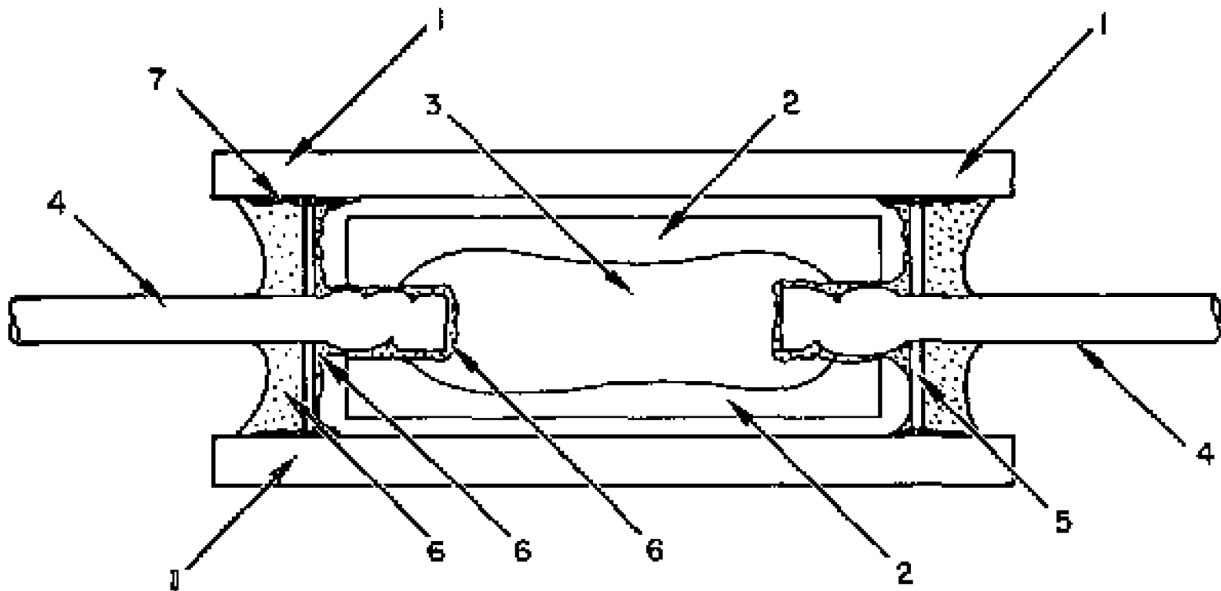
Back in 1952, carbon composition resistors were not held in particularly high regard:

*Such resistors are not suitable for precision work because of their poor stability, high temperature coefficient, variation of resistance with applied voltage, poor performance at very high frequencies, etc.*<sup>2</sup>

Despite their poor reviews at the time, carbon composition resistors were widely used in all types of electronic equipment, including guitar amplifiers. In 1953 the Radiotron Designer's Handbook offered these suggestions for reducing the noise that they add to the signal path:<sup>3</sup>

*Noise in the plate load resistor (current noise) may be avoided by using some form of special low-noise resistor, such as a high-stability cracked carbon resistor ... Composition resistors may be used as screen resistors with pentodes, because the noise voltage is bypassed to earth. Current noise, however, occurs in the grid circuit due to the negative grid current of the valve. Cathode bias resistors (if used) should be wire wound.*

A carbon-composition-rod resistor consists of a solid rod formed by a mixture of fine carbon particles and a binding medium. Its resistance value depends on the length and diameter of the rod and on the conductivity of the mixture. Here are the main features of a carbon composition resistor manufactured in 1963.4



1. outer case, fused ceramic tube with a high percentage of magnesium and silica and a low percentage of aluminum and iron
2. inner case, phenolic plastic impregnated with silicon oxide
3. carbon composition element, organic resin mixed with carbon granules
4. lead wire, tin-coated copper
5. nickel washer
6. solder made from tin, lead, and silver
7. gold

### Alternative Types of Resistors

Film resistors are made of a cylindrical core covered by a film. A spiral is cut into the film to vary the resistance, which can be made very precise by measuring continuously as the spiral is cut, then stopping when the desired value is reached. A cracked carbon resistor is made by cracking a hydrocarbon over a non-conducting rod while in an oxygen-free atmosphere. The resistance is based on the thickness of the film and on the amount of carbon and contaminants that it contains.

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Metal film resistors are made from mixtures of nickel and other materials like chromium, copper, silicates, or phosphorous. The films are usually created by physical vapor deposition.

Metal glaze or Cermet resistors are often used for high ohmic values. They are manufactured by dipping a non-conducting ceramic rod in a paste and then drying and firing the resulting film. With this process, stable resistors with values as high as 100MΩ can be produced.

### Noise Specific to Carbon Composition Resistors

*Current noise*, which is particularly significant for carbon composition resistors at audio frequencies and below, was studied as far back as 1919 in an unpublished report by T.S. Kawamoto written for the Western Electric Company.<sup>5</sup> In 1955, George Conrad made measurements at 1kHz that demonstrated that current noise power in carbon composition resistors can be 1000 times greater than [thermal noise](#).<sup>6</sup> This is an important consideration for an audio circuit that uses these resistors, especially for the sensitive first stage of a preamp.

An in-depth analysis of carbon composition resistor noise and nonlinearity was conducted in 1958 and published by T.R. Williams and J.B. Thomas.<sup>7</sup> They affirmed what researchers before them had concluded, that the noise spectrum, measured in volts squared per hertz, is

$$S(f) = K \frac{I^\gamma}{f^\delta}$$

where  $I$  is the DC current in amps,  $f$  is the frequency in hertz,  $\gamma$  is a constant that is usually close to 2, and  $\delta$  is a constant close to 1. The factor  $K$  is a constant for a particular resistor pulled out of a parts bin, but it varies considerably, even for the same manufacturer, model number, and resistance value.

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For a half-watt resistor operating at a dissipation of 50 milliwatts or more, as often is the case for the plate load resistor in a preamp voltage amplification stage, noise does not increase with current quite as much, so  $\gamma$  decreases somewhat from its value at very low current levels. The important conclusion that we draw from looking at the equation is that the noise increases with DC current and decreases with frequency.

Terman and Pettit note that the noise level in carbon composition resistors varies greatly from resistor to resistor, with variations of 20 to 1 being typical in a group of otherwise identical resistors.<sup>8</sup> This is reflected by a large range of possible values for the factor  $K$ .

Williams and Thomas determined that in addition to the wide variance in noise level between individual resistors, the noise level of a single resistor has large random jumps over time, both increasing and decreasing, especially for the first several minutes after DC current has been applied. For some of the resistors in their study, the jumps were so large that it was impossible to make consistent noise measurements. Ten years later, Campbell and Chipman reported this same behavior at 20kHz and above.<sup>9</sup> For a guitar amp builder, this means that a particularly noisy carbon composition resistor can likely be fixed by simply replacing it, even if the replacement is an identical part.

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## What does carbon composition resistor noise sound like?

Conrad, who worked for the National Bureau of Standards, provides a vivid description of current noise:

*The sound of amplified current-noise heard with earphones is often distinguishable from the hiss-like sounds of thermal noise, flicker noise, etc. With some resistors the erratic sputtering sound predominates, whereas with others, even though they may be "identical manufacturer's" samples, erratic sounds may be entirely missing with only the more familiar hiss evident. Erratic sounds are usually most noticeable immediately after initiation of the DC current. In general, the more noisy the resistor, the larger the content of the erratic sounds. Occasionally, samples are found in which only steady crashing sounds can be heard.*

He also notes that "wild" samples produce a noise spectrum far beyond what is expected. Fortunately, however, he did not find very many resistors that fit this category.

## References

1Dave Hunter, **The Guitar Amp Handbook**, updated and expanded Ed., (Milwaukee: Backbeat Books, 2015), p. 254.

2Frederick Terman and Joseph Pettit, **Electronic Measurements**, 2nd Ed., (New York: McGraw-Hill, 1952), p. 632.

3F. Langford-Smith, ed., **Radiotron Designer's Handbook**, 4th ed., (Harrison: RCA, 1953), p. 783.

4Herbert Y. Tada, "Aspects Affecting the Reliability of a Carbon Composition Resistor," **IEEE Transactions on Component Parts**, Vol. 10, No. 2, June 1963, pp. 67-79.

5R.H. Campbell, Jr. and R.A. Chipman, "Noise from Current-Carrying Resistors 20 to 500 Kc," **Proceedings of the IRE - Waves and Electrons Section**, Vol.37, No. 8, August 1949, pp. 938-942.

6George T. Conrad, Jr. "Noise Measurements of Composition Resistors, Part II," **Transactions of the IRE Professional Group on Component Parts**, Vol. 4, No. 1, November 1955, pp. 61-78.

7T.R. Williams and J.B. Thomas, "Current Noise and Nonlinearity in Carbon Resistors," **IRE Transactions on Component Parts**, Vol. 5, No. 4, December 1958, pp. 151-153.

8Frederick Terman and Joseph Pettit, **Electronic Measurements**, 2nd Ed., (New York: McGraw-Hill, 1952), p. 635.

9R.H. Campbell, Jr. and R.A. Chipman, "Noise from Current-Carrying Resistors 20 to 500 Kc," **Proceedings of the IRE - Waves and Electrons Section**, Vol.37, No. 8, August 1949, pp. 938-942.